The One-Nucleon Universe: Simulating Hadronic Benevolence via Neutron Architecture

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Neutron n | Mass: 939.565 MeV | Lifetime: 878.4 s | Spin: 1/2 | Charge: 0 | Quark content: udd **Source:** Extends `v1.0Neutron Deep Dive_ One-Nucleon Universe.pdf` (V1.0)

Δ Change Log V1.0 □ V2.0

Parameter	V1.0	V2.0	Source
Neutron lifetime puzzle	$\tau_n \sim 880$ s noted	V2.0: Beam vs. bottle tension persists at 3.4σ. Bottle: $\tau_n = 877.75 \pm 0.34$ s. Beam: 888.0 ± 2.0 s. Dark decay hypothesis: n□$\chi+\gamma$ with BR~1% could explain gap. V2.0: adds DARK_NEUTRON_DECAY branching ratio as configurable parameter	UCNτ 2023; Beam lifetime 2022
Neutron EDM search	nEDM limit noted	V2.0: nEDM Collaboration (2020, confirmed 2023):	d_n
Neutron star merger GW constraints	Not covered	V2.0: GW170817 + GW230529 (LIGO O4): neutron star radius $R = 11.7 \pm 1.0$ km. Tidal deformability Lambda=190 ± 120. Constrains nuclear matter equation of state. V2.0: adds NEUTRON_STAR_MATTER mode where density > 2×nuclear saturation density	LIGO GW170817 2017; GW230529 2024

Parameter	V1.0	V2.0	Source
Free neutron quantum reflection	Not covered	V2.0: GRANIT (ILL) 2023: neutrons in gravitational quantum states E_n = (m_n^2 g^2 hbar$^{2/2)}$(1/3) × z_n. First 3 quantum states measured. V2.0: implements quantum gravitational orbital states for ultra-cold neutrons in 'cosmic level' simulation	GRANIT ILL 2023

---

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the "Origami Universe" metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,          // Spinor components
    berry_phase: f32,     // Geometric phase accumulated
    proper_time: f32,     // tau - worldline parameter
    color_charge: u32,    // Packed color in 2 bits (R=0,G=1,B=2)
    quantum_numbers: u32, // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step
```

```
    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}
```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m_e	$9.10938356 \times 10^{-31}$ kg	$9.1093837139 \times 10^{-31}$ kg
e	$1.60217663 \times 10^{-19}$ C	$1.602176634 \times 10^{-19}$ C (exact)
hbar	$1.054571817 \times 10^{-34}$ J·s	$1.054571817 \times 10^{-34}$ J·s (exact)
c	2.99792458×10^{8} m/s	2.99792458×10^{8} m/s (exact)
alpha	1/137.036	1/137.035999084 ±0.000000021

---

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```
// V2.0 Universal Particle State — extends V1.0
struct ParticleInstanceV2 {
    // □□ Inherited from V1.0 □□□□□□□□□□□□□□□□□□□□□□□□□□
    double proper_time;           // tau worldline parameter (meter)
    glm::dvec3 position;          // 3-space vector
    glm::dvec3 momentum;          // 3-momentum (relativistic)
    float spin_z;                 // +0.5 or -0.5 (or eigenvalue)
    float berry_phase;            // Accumulated Berry/Aharonov-Bohm phase

    // □□ V2.0 ADDITIONS □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□
    float entanglement_r;         // Reduced density matrix purity 0-1
    int entanglement_partner_id; // -1 = none; global particle ID
    float flavor_oscillation_phase; // For neutrinos/kaons
    uint32_t color_bits;          // 3-bit: R(b0),G(b1),B(b2)
    float decay_probability_dt;   // P(decay in dt) for unstable particles
    float vacuum_alignment;       // Order parameter for condensate mode

    // □□ WebGPU Buffer packing □□□□□□□□□□□□□□□□□□□□□□□□□
    // Total: ~100 bytes/particle □ 10^6 particles = 100 MB GPU memory
    float _pad[2];                // Align to 16 bytes for WGSL
};
```

---

8. V2.0 Benevolence Metric Update

The "Benevolence" concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

Benevolence = **Quantum Coherence** + **Information Preservation** + **Symmetry Maintenance**

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - C(t) = l_1-norm coherence of particle density matrix (Baumgratz et al. 2014) - F(t) = quantum Fisher information / classical Fisher information (ratio □ 1) - S(t) = fraction of symmetries unbroken at time t

V2.0 game mechanic: Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

End of v1.0Neutron Deep Dive_ One-Nucleon Universe V2.0 | Tardigradia Game Engine SubParticles Series This document supersedes V1.0 in all physics specifications.